

SIMATS ENGINEERING
C02-AT2
DEBUGGING
CSA0310-DATA STRUCTURE
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1.OUTPUT

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int data) {
9     if (top == MAX - 1)
10        printf("Stack Overflow");
11    else
12        stack[++top] = data;
13 }
14
15 int main() {
16     push(10);
17     push(20);
18
19     printf("%d", stack[top]);
20
21     return 0;
22 }
```

 **Submitted**

Test Input

stdin input (optional)

Output

20

 Pending manual review

2.OUTPUT

Operation
1 mark

Q2 
5. Buggy Code in Stack POP
Operation
1 mark

Q3 
6. Buggy Code in underflow
operation
1 mark

3/3 answered

```
#define MAX 5
int stack[MAX];
int top = 2;
int pop() {
    if (top == -1)
        return -1;
    return stack[++top];
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX] = {10, 20, 30};
6 int top = 2;
7
8 int pop() {
9     if (top == -1)
10        return -1;
11    return stack[top--];
12 }
13
14 int main() {
15     printf("%d", pop());
16     return 0;
17 }
```

Test Input

stdin input (optional)

Output

30

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3.OUTPUT

4. Buggy Code in Stack Push Operation
1 mark

Q2
5. Buggy Code in Stack POP Operation
1 mark

Q3
6. Buggy Code in underflow operation
1 mark

3/3 answered

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 int pop() {
9     if (top == -1) {
10         printf("Underflow");
11         return -1;
12     }
13     return stack[top--];
14 }
15
16 int main() {
17     pop();
18     return 0;
19 }
```

Test Input

stdin input (optional)

Output

Underflow

🚫 Pending manual review